

TARCU, Romania

WMO: 153170

Lat: 45.2810N

Lon: 22.5328E

Elev: 7158

StdP: 11.27

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	2.1	-23.3	2.0	19.6	-15.5	3.1	15.6	76.1	13.8	63.7	11.8	27.7	20	1.037

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	62.0	54.1	59.4	52.3	57.2	51.3	56.1	59.8	54.3	57.7	52.8	55.9	12.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.7	83.3	57.5	52.8	77.7	55.9	51.3	73.4	54.4	27.1	60.4	25.8	57.7	24.7	56.1	63.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 63.1	54.1	45.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.1	17.6	20.0	21.6	28.6	38.1	45.7	48.0	49.6	41.2	35.5	28.5	21.2		
	DBStd	13.48	8.46	7.62	7.57	9.12	6.28	6.26	6.00	5.92	7.83	8.83	8.76	8.26		
	HDD50	6345	1004	839	881	641	369	156	111	78	276	453	644	893		
	HDD65	11660	1469	1259	1346	1091	833	579	528	476	713	914	1094	1358		
	CDD50	160	0	0	0	0	2	27	48	67	12	4	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	18.5	20.0	22.1	20.1	19.3	16.9	15.0	16.3	13.5	17.4	17.1	20.7	23.9		
	PrecAvg	33.6	2.0	1.9	1.9	2.5	3.4	4.1	4.4	3.0	3.5	2.6	2.1	2.2		
	PrecMax	50.2	4.3	6.6	4.5	4.2	6.5	8.9	9.8	8.1	7.6	5.4	5.9	4.7		
(15) Precipitation	PrecMin	23.2	0.7	0.3	0.4	0.4	1.3	1.0	1.5	0.2	0.2	0.1	0.0	0.2		
	PrecStd	8.0	0.9	1.3	1.0	0.9	1.4	1.9	2.0	1.9	1.9	1.4	1.4	1.1		
	DB	36.7	38.1	40.6	49.0	54.9	61.8	64.9	65.4	62.4	56.5	50.2	44.9			
	MCWB	25.2	31.6	35.2	42.6	48.1	54.6	55.6	55.0	56.4	41.6	36.9	30.1			
	2% DB	33.5	34.5	36.4	45.9	51.4	58.7	61.4	62.4	62.4	57.4	53.3	45.7	38.2		
(21) Mean Coincident Wet Bulb Temperatures	MCWB	25.7	29.3	32.6	39.0	46.5	52.4	54.0	53.8	51.0	41.3	35.8	28.9			
	5% DB	30.6	31.8	33.8	42.8	48.7	56.7	58.7	59.7	53.6	50.3	42.0	33.7			
	MCWB	27.7	28.7	30.5	37.8	44.9	51.3	52.3	52.3	48.6	40.6	34.4	27.8			
	10% DB	28.4	29.7	31.6	40.5	46.5	54.4	56.1	57.1	50.5	47.1	38.7	31.2			
	MCWB	25.7	27.7	29.5	36.8	43.3	50.3	51.5	51.7	45.6	40.1	34.8	27.5			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	31.3	34.1	36.0	44.0	50.1	58.0	59.0	58.0	55.9	48.0	40.7	34.1			
	MCDB	31.7	35.5	39.3	47.7	52.5	60.1	62.4	62.2	61.8	51.2	43.2	34.3			
	2% WB	30.0	31.9	33.7	41.0	47.5	54.9	55.6	55.8	51.9	45.4	38.5	31.4			
	MCDB	30.6	34.0	35.7	43.7	50.1	57.0	59.4	60.0	56.1	49.3	40.9	33.9			
	5% WB	28.0	30.1	31.8	39.0	45.5	52.8	54.0	54.0	49.1	43.3	37.4	29.7			
(31) Mean Daily Temperature Range	MCDB	29.0	30.9	33.4	41.9	47.9	55.3	56.9	57.9	52.7	47.3	40.4	32.9			
	10% WB	26.2	28.3	29.9	37.2	43.7	50.9	52.3	52.6	47.1	41.5	35.8	28.3			
	MCDB	27.7	28.9	31.2	40.4	45.8	53.2	55.0	56.1	49.6	45.5	38.2	31.3			
	5% DB	8.7	8.7	8.2	8.2	7.2	8.1	8.1	7.6	7.4	8.2	8.0	8.5			
	MCDBR	9.9	9.0	8.3	9.7	10.1	10.6	11.0	9.4	10.1	9.8	11.0	9.5			
(36) Clear-Sky Solar Irradiance	MCWBR	7.3	7.0	6.9	7.9	9.4	10.7	9.8	9.3	8.4	9.2	7.8	5.6			
	5% WB	8.3	8.3	8.6	8.9	9.0	9.9	9.2	8.7	8.4	9.1	8.1	8.0			
	MCDBR	7.8	7.8	7.6	7.8	9.5	10.6	9.8	9.0	8.1	9.4	7.0	7.1			
	taub	0.247	0.268	0.299	0.340	0.332	0.326	0.336	0.346	0.320	0.293	0.272	0.249			
	taud	2.439	2.396	2.413	2.327	2.381	2.415	2.376	2.346	2.417	2.509	2.527	2.486			
(42) All-Sky Solar Radiation	Ebn at Noon	282	294	298	292	297	298	293	286	286	281	271	271			
	Edh at Noon	22	28	32	38	37	36	37	37	31	25	20	19			
	RadAvg	421	672	1030	1391	1661	1870	1849	1673	1190	820	482	349			
(45)	RadStd	54	87	143	156	151	178	149	165	121	98	62	47			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	+1.19	N/A	N/A	N/A	N/A	N/A	-207	-315	+212	+108

Nomenclature: See separate page